CHAPTER 2

Lecture 5

Stochastic differential equations on Hilbert spaces

2.2. Stochastic differential equations with additive noise

In the particular case where the dispersion coefficient C does not depend on the solution, equation (2.1) is called a stochastic differential equation with additive noise. This case is very close to the deterministic analogue. Indeed, let

$$W_A(t) := \int_0^t T_{t-s} C \, dW_s \,, \qquad t \in [0,T] \,,$$

be the stochastic convolution and suppose that $(W_A(t))_{t\in[0,T]}$ has a version with continuous trajectories in H. Decomposing the mild solution

$$X_t = Y_t + W_A(t)$$

we formally obtain the following equation

$$dY_t = [AY_t + B(Y_t + W_A(t))] \, dt \,, Y_0 = \xi \tag{2.1}$$

for $(Y_t)_{t\in[0,T]}$. Equation (1.1) can be seen as a deterministic evolution equation with time-dependent random coefficients

$$B(\cdot + W_A(t)) \,.$$

In particular, if (1.1) has a unique mild solution $(Y_t(\omega))_{t\in[0,T]}$ for P-a.e. ω and the dependence of $(Y_t)_{t\in[0,T]}$ on ω is predictable, then $X_t = Y_t + W_A(t)$, $t \in [0,T]$, is a mild solution of (2.1).

2D-Stochastic Navier Stokes equations with additive noise

Let $D \subset \mathbb{R}^2$ be a bounded open domain with regular boundary ∂D. We consider the following stochastic Navier-Stokes equations

$$\begin{cases} \partial_t u(t,x) - \nu \Delta u(t,x) + (u(t,x)\cdot\nabla)u(t,x) + \nabla p(t,x) &= \dot{\xi}_t(x) \quad t\in[0,T], x\in D \\ \text{div } u(t,x) &= 0 \quad t\in[0,T], x\in D \\ u(t,x) &= 0 \quad t\in[0,T], x\in \partial D \\ u(0,x) &= u_0(x) \quad x\in D \,, \end{cases} \tag{2.2}$$

where $(\xi_t)_{t\geq 0}$ is a (cylindrical) Wiener process. Here, $u : [0,T]\times D \to \mathbb{R}^2$ is the velocity field, $\nu > 0$ the viscosity and $p : [0,T]\times D \to \mathbb{R}$ denotes the pressure. We will consider the equation in similar function spaces as for the deterministic case:

$$D_0^\infty = \left\{ u \in C_0^\infty(D;\mathbb{R}^2) \,,\ \text{div } u = 0 \right\}$$

$$H = \text{ closure of } D_0^\infty \text{ in } L^2(D;\mathbb{R}^2) \text{ w.r.t. } \|u\|_H^2 := \int_D |u|^2 \, dx$$

$$V = \text{ closure of } D_0^\infty \text{ in } L^2(D;\mathbb{R}^2) \text{ w.r.t. } \|u\|_V^2 := \int_D |Du|^2 \, dx$$

Applying the Helmholtz projection $\Pi : L^2(D,\mathbb{R}^2) \to H$ one obtains the following abstract evolution equation

$$\begin{cases} du(t) &= [Au(t) + B(u(t),u(t)) + f(t)] \, dt + C \, dW_t \\ u(0) &= u_0 \end{cases} \tag{2.3}$$

on H, where

- $A = \Pi\Delta_D$ is the Stokes operator on H
- $B : V \times V \to V'$, ${}_{V'}\langle B(u,v), w\rangle_V = -\int_D w(x)\cdot(u(x)\cdot\nabla)v(x)\,dx$.

THEOREM 2.1. *If* $(W_A(t))_{t\in[0,T]}$ *has a version in* V *with continuous trajectories, then* (1.3) *has a unique mild solution.*

PROOF. In this case, equation (1.1) can be written as

$$dY_t = [\nu AY_t + B(Y_t + W_A(t))]\,dt\,, Y_0 = u_0\,. \tag{2.4}$$

It is a classical result, that (1.4) has for ω with $t \mapsto W_A(t)(\omega)$, $[0,T] \to V$ continuous, a unique solution $Y_\cdot \in L^2([0,T];V)$, $\dot{Y}_\cdot \in L^1([0,T];V')$ satisfying also

$$\begin{aligned}\sup_{t\in[0,T]} \|Y_t\|_H^2 + \nu\int_0^T \|Y_t\|_V^2\,dt \\ \leq \exp\left(\frac{2}{\nu}\int_0^T \|W_A(t)(\omega)\|_V^2\,dt\right) \\ \left(\|u_0\|_H^2 + \frac{1}{\nu}\int_0^T \|W_A(t)(\omega)\|_H^2\|W_A(t)(\omega)\|_V^2\,dt\right).\end{aligned}$$

It can be also shown that the dependence of the unique solution $Y_\cdot$ on ω is predictable, so that $X_t = Y_t + W_A(t)$ is a mild solution of (1.3). □

REMARK 2.2. Regularity properties of the stochastic convolution W_A are well-studied (see the monographs [1,2]). The main difficulty with W_A is that it is not a martingale w.r.t. t. This does not contradict the properties of the stochastic integral, because for any $t > 0$ the process

$$W_A^{(t)}(s) = \int_0^s e^{(t-r)A}C\,dW_r\,, s \in [0,t]\,,$$

is a martingale up to time t.

References:

1 G. Da Prato, J. Zabczyk, Stochastic Equations in Infinite Dimensions, Cambridge University Press, 1992.
2 G. Da Prato, J. Zabczyk, Ergodicity for infinite dimensional systems, Cambridge University Press, 1996.